Physics Consistent Conditional Flow Matching Model as a Plasma Physics Surrogate Model

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Abstract

Physics-informed surrogate modeling for steady-state systems often struggles with data scarcity and optimization instabilities from soft penalty losses. We introduce a unified, continuous-time Conditional Flow Matching (CFM) framework for surrogate modeling and sparse data assimilation, evaluated on low-temperature plasma dynamics. By combining data-dependent stochastic interpolants with dynamic masking, the framework unifies unconditioned prediction and missing data imputation into a single objective. We replace standard velocity matching (v-prediction) with a direct state regression (y-prediction) objective, eliminating the variance-induced statistical inefficiencies for deterministic physics. For inference, the formulation supports both standard time-dependent ODE integration (CFM-t) and an autonomous regime (CFM-0) that enables measurement conditioning via anchor overwriting without boundary disruption. Global conservation laws are enforced via post-hoc hard geometric projections rather than loss regularizers. Results indicate that both CFM models achieve superior accuracy, sample efficiency under data scarcity, and raw physical consistency compared to standard regressors. Furthermore, macroscopic constraints alone yield negligible reductions in L_2 error, confirming the necessity of a data-driven generative manifold prior.

***K*eywords** Scientific Machine Learning · Surrogate Modeling · Conditional Flow Matching · Plasma Physics · Manifold Learning · Physics-Consistent Deep Learning

1 Introduction

The development of high-fidelity surrogate models is a central objective in scientific machine learning, particularly for complex, non-linear domains such as low-temperature plasma physics and fluid dynamics [1–5]. Because the state variables of these systems are tightly coupled by governing equations and conservation laws, valid physical states do not uniformly occupy the ambient space; rather, they are strictly confined to a lower-dimensional, highly structured data manifold $\mathcal{M}$ [3, 6]. Standard deep learning regression techniques fundamentally struggle with this geometric reality. By deterministically mapping operational inputs to outputs, they lack the probabilistic framework required to navigate the ambient space, frequently producing predictions that fall off the manifold and violate underlying physical laws [7–9]

In practical engineering workflows, generating high-fidelity data via first-principles simulators (e.g. kinetic, particle-in-cell, or chemistry solvers [10–12]) is computationally prohibitive, often yielding scarce or incomplete datasets [13]. Consequently, surrogate models cannot serve merely as static maps; they must dynamically handle both zero-shot forecasting of the full physical state and data assimilation from sparse experimental diagnostics [2, 14]. This duality unlocks important capabilities in computational physics. In multi-scale frameworks, for instance, a well-calibrated surrogate can rapidly reconstruct full, high-dimensional physical states (e.g. detailed species and vibrational densities in low-temperature plasmas) directly from coarse-grained macroscopic descriptions, bypassing the need for expensive, concurrent kinetic solvers at every grid point [3, 15–17]. Concurrently, in model-based optimization, differentiable surrogates that natively accommodate missing variables can seamlessly assimilate runtime diagnostics to continuously update their accuracy and rapidly explore vast design spaces [18, 19].

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While Physics-Informed Neural Networks (PINNs) have been widely adopted to mitigate physical inconsistencies by embedding analytical governing equations into the loss function, they frequently encounter severe optimization challenges [1, 20]. Specifically, the composite loss functions in PINNs create well-documented gradient pathologies, where the data-driven and physics-informed gradients actively conflict and fail to navigate tightly constrained manifolds [21–24]. More critically, PINNs require the physical system to be explicitly defined analytically and fully differentiable via automatic differentiation. In many complex scientific domains, the underlying simulators act as non-differentiable black boxes, rely on discrete empirical lookup tables, or involve complex legacy codes, rendering the exact computation of physical loss gradients impossible [3, 13]. In these scenarios, practitioners frequently only have access to macroscopic global conservation equations (e.g. charge neutrality, mass conservation) rather than the full closed-form dynamics [3]. Treating these broad global invariants as soft losses is geometrically weak and vastly insufficient to uniquely constrain the network to the true physical manifold [8, 9].

Recent advances in generative modeling have been heavily driven by continuous-time frameworks, most notably Diffusion and Flow Matching models [25–28]. By learning complex data distributions via stochastic or ordinary differential equations (SDEs/ODEs), these methods have achieved unprecedented success across computer vision, high-fidelity video generation, and robotics [25, 26, 29–31].

Motivated by these breakthroughs, one prominent line of recent literature has adapted diffusion models to the physical sciences. To enforce physical validity during the generative process, Bastek et al. [32] embed non-linear PDE residuals directly into the diffusion denoising objective, acting as a soft physical constraint that reduces the reliance on paired training data. Addressing complex temporal dynamics, Wei et al. [33] developed an evolving conditional diffusion framework specifically designed to satisfy time-varying boundary conditions. Concurrently, efforts in solving inverse problems and performing large-scale physical downscaling have heavily adopted continuous-time priors. Furthermore, in generative modeling, continuous-time priors have established powerful techniques for inpainting and missing data imputation [34]. Leveraging this capability, Hong et al. [35] and Jacobsen et al. [36] employ conditional score-based models to seamlessly reconstruct full physical fields from sparse sensor measurements.

To bypass the heavy sampling overhead of diffusion models, a parallel line of research has shifted toward Flow Matching. By leveraging Optimal Transport (OT) probability paths [27, 37], Flow Matching induces efficient, straight-line vector fields highly suited for deterministic physical systems. Exploiting this, Baldan et al. [38] developed a Pareto-optimal framework that dynamically balances empirical data matching with governing PDE constraints. To further accelerate inference, Bai et al. [39] introduced Spatio-Temporal MeanFlow, which replaces the abstract generative velocity field with an explicit PDE operator, enabling rapid, single-step physical integration. Finally, extending these continuous formulations to data assimilation, flow matching has been successfully adapted for inverse problems [40, 41]. These approaches demonstrate that directly regressing optimal conditional vector fields yields significantly faster and more stable reconstructions from sparse measurements than traditional diffusion.

While recent physics-informed generative models significantly accelerate continuous-time integration, their applicability is restricted to systems governed by explicit, differentiable equations. To address steady-state physical systems where such equations are mathematically or computationally inaccessible, we introduce a physics-consistent Conditional Flow Matching (CFM) surrogate, demonstrated on low-temperature plasmas [3]. By fusing continuous-time CFM [27, 28, 40, 42] with manifold representation learning [43, 44], we substitute standard independent noise with data-dependent stochastic interpolants [40] governed by a dynamic masking operator. This architectural shift seamlessly unifies zero-shot surrogate modeling and sparse data assimilation into a single continuous-time objective.

Crucially, we adapt Flow Matching to the deterministic nature of steady-state physics by replacing standard velocity matching with a direct y-prediction parameterization [45–47]. By targeting the variance-free physical state rather than highly stochastic transport vectors, the continuous flow acts as a generative extension of masked denoising autoencoders [44, 47, 48]. This objective forces the network to learn a pure geometric projection onto the low-dimensional physical manifold, allowing it to bypass the statistical inefficiencies of standard velocity regression [47].

During inference, this variance-free formulation enables flexible solution extraction through either explicit time-dependent ODE integration or an autonomous flow regime grounded in Beckmann Transport Models [49]. Finally, rather than corrupting the generative gradients with soft physics-informed penalty losses, which often cause optimization instabilities, we guarantee adherence to global conservation laws (e.g. charge neutrality and mass conservation) through post-hoc hard geometric projections [3].

The remainder of this paper is structured as follows. Section 2 establishes the theoretical background connecting regularized autoencoders, manifold learning, and continuous-time generative flows. Section 3 introduces our data-dependent interpolant, formalizes the direct y-prediction objective, and details the explicit and autonomous inference regimes alongside post-hoc physics projections. Section 4 details the experimental setup, baseline models, and empirical findings across surrogate modeling, data scarcity, and missing data imputation tasks. Finally, Section 5 provides concluding remarks and outlines directions for future work.

2 Background

2.1 The Physical Manifold and System Sloppiness

In the context of deterministic physical simulators, such as those modeling low-temperature plasma (LTP) physics, the physical state $y \in \mathbb{R}^D$ is determined by a set of experimental control inputs $x \in \mathcal{X}$ and a fixed set of physical hyperparameters $\theta \in \Theta$, where Θ denotes the domain of all physically admissible parameters characterizing the model [4, 5, 50]. Formally, the system state is defined by the steady-state fixed point of a complex, non-linear algebraic system:

$$F(x, y; \theta) = 0. \tag{1}$$

Because the simulator is deterministic for a given model parameterization θ, the conditional probability distribution of the physical state given the inputs, $p_d(y|x)$, is effectively a Dirac delta distribution centered at the solution y. Geometrically, sweeping across the bounded operational domain of inputs $\mathcal{X}$, the collection of all valid steady-state solutions maps out the exact physical data manifold:

$$\mathcal{M}_\theta = \{y \in \mathbb{R}^D \mid \exists x \in \mathcal{X} \text{ such that } F(x, y; \theta) = 0\} \subset \mathbb{R}^D. \tag{2}$$

Understanding the geometry of $\mathcal{M}$ also requires analyzing the sensitivity of the system to its underlying parameters θ. Plasma chemsitry models are usually characterized as *sloppy* [6, 50, 51]. While they rely on high-dimensional parameter spaces, their macroscopic behavior is strictly controlled by a relatively small number of stiff parameter modes. The vast majority of parameter directions are unconstrained and have minimal effect on the system's output. Analyzing the Fisher Information Matrix (FIM) with respect to θ reveals that the eigenvalues decay log-linearly, establishing that the model possesses a remarkably low effective dimensionality [6, 51].

Crucially, this sloppiness in the θ-parameter space translates directly into the geometric topology of the observation space y. The system is only sensitive to a few stiff parameter dimensions, and the model's valid predictions do not isotropically fill the ambient observation space $\mathbb{R}^D$. Instead, they are tightly confined to a bounded, lower-dimensional subspace. Information geometry describes this manifold as a *hyperribbon*, a structure characterized by an exponential hierarchy of widths and a flat extrinsic curvature along its dominant axes [6, 51].

Because F is a highly non-linear and we have a sloppy system, the mapping from experimental conditions x to the system observables y is rarely well-conditioned and globally injective. Standard neural network regression models attempt to learn a direct mapping $f(x) \approx y$ by minimizing point-wise distances in the ambient space $\mathbb{R}^D$. This approach is geometrically blind; it interpolates across non-linearities and fails to separate the meaningful physical signals from the unidentifiable degrees of freedom. Consequently, regression predictions easily fall off the hyperribbon, yielding states that violate the governing equations $F(x, y; \theta) = 0$.

To guarantee physically consistent surrogate predictions, the objective cannot simply be point-wise regression. Instead, the model must be trained to explicitly capture the intrinsic, low-dimensional geometric support of the data manifold $\mathcal{M}$ conditioned on x.

2.2 Manifold Learning via Denoising and Masked Autoencoders

To explicitly constrain the surrogate model to the underlying data manifold $\mathcal{M}$, we adopt the principles of representation learning. Standard generative implementations often target the abstract prediction of auxiliary noise (ϵ-prediction) [25, 26]. In contrast, we design our framework to capture the intrinsic geometry of the physical manifold itself by learning a restorative vector field that maps degraded states back to valid physical solutions [43, 47, 48].

This training procedure directly mirrors the manifold assumption through the lens of self-supervised reconstruction. We begin with a valid physical state $y \in \mathcal{M}$ and intentionally apply a stochastic degradation process, producing a corrupted observation $\tilde{y} \sim q(\cdot|y)$. Geometrically, this degradation displaces the data point off the valid physical manifold and into the surrounding ambient space. By formulating this degradation as a generalized process q, we can encapsulate two autoencoding paradigms within a single formulation. When the corruption is modeled as additive isotropic Gaussian noise, defined as $q(\tilde{y}|y) = \mathcal{N}(\tilde{y}; y, \sigma^2\mathbf{I})$ the framework operates as a continuous Denoising Autoencoder (DAE) [48]. Alternatively, when the degradation is applied as a discrete structural corruption via a stochastic binary mask, $\tilde{y} = S \odot y$, it functions as a Masked Autoencoder (MAE) [44].

In systems such as plasma chemistry, the state vector y possesses a high signal-to-noise ratio [52]. Unlike natural images where adjacent pixels often contain redundant information, every individual component of a physical state vector (e.g. specific species densities, electron temperatures) carries non-redundant physical meaning. Furthermore, these variables are tightly coupled by underlying governing and conservation laws. Consequently, forcing a neural network to reverse structural masking or continuous noise necessitates more than superficial pattern recognition; it forces the model to explicitly learn and exploit these fundamental physical correlations. By inferring the missing or corrupted variables from the available conditional context, the network *learns* the geometry and governing constraints of the physical manifold [48, 53, 54].

The neural network g_ϕ, parameterized by weights ϕ, is tasked with reversing this generalized degradation, conditioned on the physical input x. Let $q(\cdot|y)$ define the chosen forward stochastic corruption distribution. By minimizing the expected reconstruction error between the network's prediction $g_\phi(\tilde{y}, x)$ and the clean signal y:

$$\mathcal{L}(\phi) = \mathbb{E}_{(x,y)\sim p_d} \mathbb{E}_{\tilde{y}\sim q(\cdot|y)} \left[\|g_\phi(\tilde{y}, x) - y\|_2^2 \right], \tag{3}$$

where p_d is the empirical data distribution, the model is trained to reverse the corruption process.

Crucially, the theory of regularized autoencoders shows that minimizing this objective forces the network to learn a restorative vector field, projecting corrupted points from the ambient space directly back onto the intrinsic data manifold [48,55]. This formulation also establishes a natural bridge to generative modeling: under Gaussian degradation, the optimal denoiser g_{ϕ^*} implicitly learns the score function (the gradient of the log-density) of the underlying data distribution [43].

2.3 Generative Modeling via Continuous Normalizing Flows and Flow Matching

To learn the geometric structure of the physical manifold, we frame the surrogate task using the framework of Continuous Normalizing Flows (CNFs) [56]. CNFs construct a continuous-time diffeomorphic map, or flow, denoted as $\phi_t : \mathbb{R}^d \to \mathbb{R}^d$, that transforms a simple prior probability density $p_0(x)$ (e.g. standard Gaussian noise) into a complex target data distribution $p_1(x) \approx q(x)$. This transformation is governed by a time-dependent vector field $v_t : [0, 1] \times \mathbb{R}^d \to \mathbb{R}^d$, which defines an Ordinary Differential Equation (ODE):

$$\frac{d}{dt}\phi_t(x) = v_t(\phi_t(x)); \quad \phi_0(x) = x. \tag{4}$$

A vector field v_t generates a probability density path p_t if it satisfies the continuity equation. The goal of generative modeling in this context is to approximate the true, unknown vector field $u_t(x)$ that generates the desired probability path from p_0 to p_1. To achieve this, the Flow Matching (FM) objective proposes a simulation-free approach to regress a neural network parameterized vector field $v_\theta(x, t)$ directly onto the target vector field $u_t(x)$ [27, 28, 57]:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,p_t(x)} \|v_\theta(x, t) - u_t(x)\|^2. \tag{5}$$

Since marginalizing over the entire empirical dataset renders the objective intractable, Flow Matching relies on a per-sample formulation. It defines conditional probability paths by constructing a time-dependent stochastic interpolant I_t that explicitly connects a base sample $x_0 \sim p_0$ to a target sample $x_1 \sim p_1$. This allows for the construction of a tractable conditional Flow Matching objective [27]:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,q(x_1),p_t(x|x_1)} \|v_\theta(I_t, t) - u_t(I_t|x_1)\|^2. \tag{6}$$

Crucially, optimizing this conditional objective provides identical expected gradients with respect to θ as the intractable marginal FM objective [27, 28].

Despite its scalability, a fundamental challenge in standard FM is the choice of coupling between the base and target distributions. Standard formulations construct the interpolant I_t using an independent coupling, where x_0 and x_1 are drawn independently. However, independent coupling inherently produces intersecting linear interpolants [27, 37, 42]. When interpolant trajectories intersect in the state space, the learned marginal vector field is forced to average incompatible directions. This geometric conflict limits the model's efficiency, resulting in curved generative paths, training instability, and higher transport costs [37, 58].

3 Methodology

3.1 Data-Dependent Interpolant and Task Unification

To formalize the generative transport from the ambient space to the target physical state $x_1 = y \in \mathcal{M}$, we construct a continuous-time stochastic interpolant I_t. To resolve the geometric conflicts of independent coupling, we employ a data-dependent coupling [40]. Under this paradigm, samples from the base density x_0 are computed conditionally given samples from the target density x_1, as it enables the construction of highly efficient dynamical transport maps that act as conditional generative models.

We define the primary interpolant path by perturbing a target-dependent base state x_0 with a time-modulated schedule:

$$I_t = \alpha_t x_0 + \beta_t x_1, \tag{7}$$

where $t \in [0, 1]$ and α_t, β_t are scalar schedule coefficients [27, 28, 40]. Following the data-dependent coupling framework introduced by Albergo et al. [40], we construct the base state x_0 to incorporate a binary observation masking operator $S \in \{0, 1\}^D$ (sampled from a distribution $S \sim p(S)$) alongside a configurable noise-gating operator $M_\epsilon \in \{0, 1\}^D$:

$$\begin{aligned} x_1 &= y, \\ x_0 &= S \odot y + M_\epsilon \odot (\sigma\epsilon), \end{aligned} \tag{8}$$

where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is standard Gaussian noise and σ defines the ambient noise amplitude, and $\odot$ denotes the Hadamard (element-wise) product. By treating S as a dynamic variable during training, this formulation establishes a unified framework capable of addressing two distinct scientific workflows within a single neural network architecture:

Forward Surrogate Modeling: When no variables are observed prior to generation, the mask evaluates to $S = \mathbf{0}$. The base state collapses to $x_0 = M_\epsilon \odot (\sigma\epsilon)$, requiring the network to map the entire state from pure ambient noise. This trains the model as a zero-shot surrogate.

Missing Value Imputation: When partial sensor measurements are available ($S \succ \mathbf{0}$), the observed variables $S \odot y$ provide a structural anchor. The model restricts transport to the unobserved subspace while keeping known coordinates fixed, unifying continuous denoising and masked autoencoding to reconstruct the underlying physical manifold $\mathcal{M}$.

Within this unified formulation, the exact generative dynamics are governed by the choice of the schedule coefficients (α_t, β_t) and the noise-gating operator (M_ϵ). Specifically, M_ϵ dictates a trade-off between ambient space regularization and inference-time complexity:

Global Noise ($M_\epsilon = \mathbf{1}$): Applying isotropic noise globally across all dimensions ensures that the base density is well-defined over the entire higher-dimensional ambient space, effectively smoothing the data manifold and alleviating singularities inherent to lower-dimensional supports [40]. However, this global perturbation introduces an inference-time burden: because noise is present across all coordinates, preserving the integrity of observed sensor values requires evaluating the exact time-dependent noise schedule ($\sigma_t = \alpha_t\sigma$) to dynamically re-harmonize the trajectory at each step [41, 59].

Asymmetric Noise ($M_\epsilon = \mathbf{1} - S$): Restricting the noise exclusively to the unobserved subspace ($\mathbf{1} - S$) treats the observed variables as clean, deterministic boundaries throughout the entire trajectory. While this sacrifices manifold smoothing on the observed dimensions, it makes boundary harmonization trivial during inference, requiring no explicit temporal clock to evaluate σ_t.

Driven by these geometric constraints, our methodology bifurcates into two distinct paradigms: a time-dependent formulation (CFM-t) utilizing global noise, and a time-independent autonomous formulation (CFM-0) utilizing asymmetric noise and a self-stopping schedule [49, 59].

3.2 Neural Parameterization and Training Objective

We define a single neural network, $g_\phi(I_t, x, S, [t])$, designed to handle both surrogate modeling and missing value imputation. The network maps the interpolant state I_t to a target output, conditioned on the global operational parameters x and the binary mask S. The explicit temporal input t is included in the time-dependent formulation (CFM-t) but omitted in the autonomous formulation (CFM-0), forcing the latter to navigate purely via spatial geometry [49, 60].

To ensure the network generalizes across all possible data assimilation configurations alongside the zero-shot surrogate regime ($S = \mathbf{0}$), we dynamically randomize the observation mask during training via $S \sim \text{Bernoulli}(p_{\text{obs}})$, where the observation probability is drawn uniformly per batch, $p_{\text{obs}} \sim \mathcal{U}(0, 1)$ [40, 44] .

Because the stochastic interpolant defines the probability path $I_t = \alpha_t x_0 + \beta_t x_1$, the instantaneous conditional transport velocity is given by $\dot{I}_t = \dot{\alpha}_t x_0 + \dot{\beta}_t x_1$ [27, 40]. Standard Flow Matching frameworks employ direct velocity regression (v-prediction) to minimize $\mathbb{E}\left[\|v_\theta(I_t, t) - \dot{I}_t\|^2\right]$ [27, 28, 46], which recovers the optimal vector field as the conditional expectation $v^*_\theta(I_t, t) = \mathbb{E}[\dot{I}_t | I_t, t]$ [27, 40].

However, because $\dot{I}_t$ retains an explicit dependence on the Gaussian ambient perturbation ϵ, regressing v requires the network to approximate a high-variance target across intermediate time steps. Furthermore, while the clean physical data y resides on a low-dimensional manifold $\mathcal{M}$, the velocity v and noise ϵ are inherently off-manifold and occupy the full ambient space [47]. Training a network to predict v forces it to allocate substantial capacity toward modeling off-manifold noise structures, which is statistically inefficient and inflates sample complexity for deterministic systems.

To resolve these limitations, we adopt a direct clean-state parameterization (y-prediction) [47]. Targeting the variance-free physical state y allows the network to act as a pure geometric projector that filters ambient perturbations and maps corrupted states back onto the manifold $\mathcal{M}$. It casts continuous Flow Matching into an equivalent continuous-time Masked and Denoising Autoencoder objective [43,44,47,48]:

$$\mathcal{L}_y(\phi) = \mathbb{E}_{S\sim p(S)}\mathbb{E}_{t\sim\mathcal{U}(0,1)}\mathbb{E}_{(x,y)\sim p_d}\mathbb{E}_{\epsilon\sim\mathcal{N}(0,\mathbf{I})}\left[\|g_\phi(I_t, x, S, [t]) - y\|_2^2\right]. \tag{9}$$

Because the physical system is deterministic, the target exhibits zero conditional variance (Var$[y|x, S] = 0$), making it a statistically efficient learning objective.

3.3 Generative Inference

At inference, sample generation corresponds to numerically integrating the deterministic Probability Flow ODE from the base distribution $I_0 \sim p_0$ at $t = 0$ to the target physical distribution $I_1 \approx y \in \mathcal{M}$ at $t = 1$ [27,40,56]. Under the y-prediction parameterization, the trained network $g_\phi(I_t, x, S, [t])$ acts as a continuous spatial projection operator. By substituting this prediction $\hat{y}$ back into the analytical time derivative of the generalized data-dependent interpolant (Eq.(7)), we obtain the exact instantaneous transport velocity used by the numerical solver:

$$dI_t = \left[\dot{\alpha}_t\Big(S \odot \hat{y} + M_\epsilon \odot (\sigma\epsilon)\Big) + \dot{\beta}_t\hat{y}\right] dt. \tag{10}$$

The dynamics of this ODE integration (the schedule definitions α_t, β_t and noise-gating mechanisms M_ϵ) depend on the temporal parameterization of the network. We establish two distinct operational regimes.

3.3.1 Explicit Time-Dependent Flow (CFM-t)

In the time-dependent regime (CFM-t), the neural network $g_\phi(I_t, x, S, t)$ receives the integration time $t \in [0, 1]$ as an explicit conditioning variable, commonly encoded via sinusoidal or learned temporal embeddings [25,40,61]. To optimize global manifold coverage and prevent dimensional singularities during representation learning, we employ global ambient noise ($M_\epsilon = \mathbf{1}$) coupled with an OT linear schedule [27,40,47]:

$$\alpha_t = 1 - t, \quad \beta_t = t. \tag{11}$$

Sample generation is carried out by integrating the Probability Flow ODE from $t = 0$ to $t = 1$ using standard numerical integrators (e.g. Euler or Heun solvers) [45,47].

In missing value imputation tasks ($S \succ \mathbf{0}$), finite discretization steps introduce cumulative numerical drift that can perturb the observed variables away from their theoretical trajectories [59,62]. To enforce measurement consistency without altering unobserved flow dynamics, we adapt the replacement boundary conditioning technique widely used in generative inpainting and inverse problems [40,59,63,64].

Let $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ denote the fixed initial noise realization sampled at $t = 0$. After computing the intermediate state update $I^{\text{raw}}_{t+\Delta t}$ via the numerical ODE step, the observed dimensions are analytically harmonized by projecting them onto their prescribed time-dependent interpolant trajectory [40,59]:

$$I_{t+\Delta t} = S \odot \Big(y_{\text{obs}} + \sigma_{t+\Delta t}\epsilon\Big) + (\mathbf{1} - S) \odot I^{\text{raw}}_{t+\Delta t}, \tag{12}$$

where $y_{\text{obs}} = S \odot y$ denotes the clean sensor measurements. This discrete projection ensures that the observed coordinates follow an exact, drift-free path toward their unperturbed target values as $\alpha_1 \to 0$, while the unobserved coordinates evolve unconstrained along the learned vector field on the physical manifold $\mathcal{M}$.

3.3.2 Implicit Time-Independent Autonomous Flow (CFM-0)

Alternatively, we employ an autonomous parameterization where the neural network $g_\phi(I, x, S)$ is explicitly blind to the continuous integration time t. Grounded in the theory of Beckmann Transport Models (BTM) [49] and Equilibrium Matching [65], learning a time-independent velocity field $v(x) = \mathbb{E}_{t,x_0,x_1}[\dot{I}_t \mid I_t = x]$ defines a mathematically valid transport from base to target distributions provided the physical state resides on a lower-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$. Trajectories of the resulting autonomous system asymptotically converge to the physical support $\mathcal{M}$ [49].

To bypass arbitrary temporal stopping criteria during numerical integration, the continuous flow must natively self-terminate on the physical manifold. Following Cheuk-Kit et al. [49], we enforce a vanishing terminal velocity ($\dot{I}_1 = \mathbf{0}$) by deploying a self-stopping quadratic schedule:

$$\alpha_t = (1-t)^2, \quad \beta_t = 1 - (1-t)^2 = 2t - t^2. \tag{13}$$

Because $\dot{\alpha}_1 = \dot{\beta}_1 = 0$, the learned autonomous drift smoothly decays to zero on the target manifold, ensuring trajectories freeze at the physical equilibrium without overshooting.

Furthermore, evaluating continuous temporal noise schedules ($\sigma_t = \alpha_t \sigma$) is mathematically ill-posed in an autonomous architecture devoid of a temporal clock. To reconcile this with data assimilation, we pair the self-stopping schedule with the asymmetric noise operator ($M_\epsilon = \mathbf{1} - S$). This eliminates ambient perturbation from the observed subspace at initialization ($t = 0$), keeping the measurement coordinates clean throughout the flow. Consequently, inference collapses into a clock-free, iterative fixed-point update:

$$I^{(k+1)} = S \odot y_{\text{obs}} + (\mathbf{1} - S) \odot \left(I^{(k)} + \eta \cdot \left(I^{(k)} - g_\phi(I^{(k)}, x, S) \right) \right), \tag{14}$$

where $\eta > 0$ is a virtual relaxation step size.

This formulation ensures zero distributional shock when known physical anchors are assimilated, allowing the autonomous generative process to collapse into a spatial fixed-point iteration that settles exactly on the physical equilibrium state.

The pseudocode for the unified surrogate training loop (Algorithm 1), the time-dependent numerical integration (Algorithm 2), and the autonomous fixed-point iteration (Algorithm 3) is provided in Appendix C

3.4 Physics Consistency and Geometric Projection

3.4.1 Decoupling Global Conservation Laws from Manifold Learning

Enforcing physical consistency is essential in scientific surrogate modeling; however, embedding conservation laws as soft regularization penalties introduces fundamental optimization challenges [22, 66]. Global invariants (such as total mass or charge neutrality) represent necessary physical conditions, yet they remain insufficient to uniquely determine the state of a non-linear system [2, 3].

Let $y \in \mathbb{R}^D$ denote the ambient state space. The true steady-state physical manifold $\mathcal{M} \subset \mathbb{R}^D$, governed by non-linear operators $F(x, y; \theta) = 0$, is inherently low-dimensional [2, 67]. In data assimilation, sensor observations (defined by the binary mask $S \in \{0,1\}^D$) confine admissible states to an affine subspace $\mathcal{V}_{\text{obs}} = \{\hat{y} \in \mathbb{R}^D \mid S \odot \hat{y} = S \odot y\}$. Consequently, the generative model must transport an ambient prior onto a distribution supported strictly on the conditional manifold $\mathcal{M}_{\text{obs}} = \mathcal{M} \cap \mathcal{V}_{\text{obs}}$ [27, 49].

Enforcing k global physical invariants $G(y) = C$ defines a constraint hypersurface $\mathcal{C} \subset \mathbb{R}^D$ [3]. Intersecting this with observations yields the constrained subspace $\mathcal{C}_{\text{obs}} = \mathcal{C} \cap \mathcal{V}_{\text{obs}}$, with codimension at most $k + \|S\|_1$ and ambient dimension $D - k - \|S\|_1$. Although the physical manifold is a subset ($\mathcal{M}_{\text{obs}} \subset \mathcal{C}_{\text{obs}}$), its intrinsic dimension is substantially smaller, $d_{\mathcal{M}} \ll D - k - \|S\|_1$, as $D \gg k + \|S\|_1$, $\mathcal{C}_{\text{obs}}$ is a vast, underspecified region in $\mathbb{R}^D$ [47, 67]. Penalizing invariant violations via soft loss functions therefore provides an ineffective geometric bias, allowing predictions to settle arbitrarily within $\mathcal{C}_{\text{obs}}$ rather than guiding them toward $\mathcal{M}_{\text{obs}}$ [3]. As a result, to preserve generative learning fidelity, we decouple global conservation laws from training and enforce them via exact post-hoc geometric projections [3].

3.4.2 Gradient Conflict, Interpolant Distortion, and Sampling Overhead

Beyond the dimensional mismatch, soft physical constraints actively degrade generative training. Introducing a penalty term yields the composite objective:

$$\mathcal{L}(\theta) = \lambda_{\text{data}} \mathcal{L}_{\text{CFM}}(\theta) + \lambda_{\text{phys}} \|G(\hat{y}, x) - C\|_2^2, \tag{15}$$

where $\hat{y}$ is obtained either from a single-step network prediction at time t or by unrolling a multi-step ODE solver. This formulation introduces two critical failure modes.

First, it creates directional gradient conflicts during backpropagation [22, 66]. The data gradient $\nabla_\theta \mathcal{L}_{\text{CFM}}$ guides the vector field along straight-line trajectories toward the exact solution $y \in \mathcal{M}$, whereas the physics gradient $\nabla_\theta \mathcal{L}_{\text{phys}}$ pulls $\hat{y}$ toward the nearest point on $\mathcal{C}$ [38]. Because $\mathcal{C}$ is vastly higher-dimensional than $\mathcal{M}$, these gradients are frequently orthogonal or opposing [21], bending the learned generative trajectories.

Second, evaluating constraints during intermediate flow states introduces severe computational and mathematical issues. Standard CFM computes expectations over sampled time steps $t \sim \mathcal{U}(0, 1)$ [27]. Penalizing intermediate states

($t \ll 1$) with steady-state laws is mathematically ill-posed, as noise-dominated transients have not reached physical equilibrium [38, 68]. Conversely, unrolling a multi-step solver during training to reach steady state drastically inflates memory and computational overhead.

3.4.3 Hard Geometric Projection Method and Constrained Flows

To avoid optimization pathologies, global physical invariants are decoupled from generative training and treated as non-learnable, exact geometric boundaries [3, 8]. The CFM objective is dedicated to learning the underlying data manifold $\mathcal{M}$, while physical compliance is enforced via a deterministic hard projection applied strictly after generative inference reaches the steady-state prediction $\hat{y}$ [9, 69].

In data assimilation tasks, the projection must simultaneously satisfy the macroscopic invariants ($\mathcal{C}$) and preserve known observations ($\mathcal{V}_{\text{obs}}$). We define the projection operator $\Pi_{\mathcal{C}_{\text{obs}}}$ to map $\hat{y}$ to the nearest valid state $y^* \in \mathcal{C}_{\text{obs}}$ via orthogonal L_2-minimization [3]:

$$y^* = \Pi_{\mathcal{C}_{\text{obs}}}(\hat{y}) := \arg \min_{y \in \mathcal{C} \cap \mathcal{V}_{\text{obs}}} \|\hat{y} - y\|_2^2. \tag{16}$$

The implementation of $\Pi_{\mathcal{C}_{\text{obs}}}$ depends on the functional form of $G(y)$. For linear conservation laws (e.g. mass or charge neutrality), the operation resolves to an exact closed-form projection onto an affine hyperplane with the observed dimensions $S \odot y$ held fixed [69]. For non-linear constraints, the projection is resolved using standard constrained numerical optimization algorithms, treating the observed variables $S \odot y$ as fixed boundary conditions [3].

This post-hoc formulation guarantees physical compliance to machine precision without hyperparameter tuning (λ_{phys}) or training overhead. Decoupling the projection step ensures that generative vector field learning proceeds unhindered while preserving both physical validity and measurement integrity.

4 Experiments

4.1 Experimental Setup

Dataset and Preprocessing. We evaluate our formulation on the low-temperature plasma (LTP) dataset introduced by [3], generated using the LoKI simulator [10, 11, 70]. The dataset comprises 3000 steady-state samples mapping three continuous experimental control parameters (gas pressure P, discharge current I, and tube radius R, $x \in \mathbb{R}^3$) to a physical state vector $y \in \mathbb{R}^{17}$ containing chemical species densities (electrons, ions, and excited neutral states). Further physical details regarding the plasma chemistry are available in [4]. Prior to training, all input features and target variables are standardized to zero mean and unit variance using standard z-score normalization to ensure stable optimization dynamics. The default dataset is partitioned using a fixed random seed into a training set (80%), a validation set for early stopping (6%), and a holdout test set (14%) reserved exclusively for evaluation.

Model Architectures and Baselines. To ensure a fair comparison under a controlled computational budget, all neural architectures are constrained to a default comparable parameter footprint ($\approx$ 19.5k parameters). Both generative models (CFM-t and CFM-0) utilize a residual network backbone. For the time-dependent flow (CFM-t), the continuous integration time $t \in [0, 1]$ is projected via sinusoidal positional encodings [25]. This temporal embedding is merged with the physical condition x to modulate the intermediate hidden layers via Adaptive Layer Normalization (AdaLN) [71, 72]. For the autonomous flow (CFM-0), the time-embedding branch is omitted, and the AdaLN layers are modulated exclusively by x, forcing the network to learn a static, time-invariant vector field. As a deterministic baseline, we employ a standard Multi-Layer Perceptron (MLP) regressor constrained to the same parameter budget, adopting the architecture and training procedure detailed in [3]. Full architectural details and training hyperparameters are provided in Appendix A.

Inference Dynamics. CFM-t generation uses a second-order Heun solver over 100 steps (200 NFE), applying boundary harmonization at each increment. To ensure exact computational parity, the autonomous CFM-0 executes a fixed-point iteration capped at 200 steps. Because CFM-0 self-terminates upon reaching the physical manifold, this budget serves as a convergence safeguard.

Physical Consistency and Post-Hoc Projection. We assess physical validity via residual errors on three global conservation laws governing the plasma system (Table 1) [3, 4]. All models are evaluated both *raw* (direct network output) and *projected*. Following the procedure of [3], we solve the projection problem using non-linear interior-point optimization (Ipopt via CasADi [73]) to machine precision, locking the known experimental observations $S \odot y$.

Table 1: Fundamental global conservation laws governing the steady-state plasma system.

Conservation Law	Equation	Physical Description
Ideal Gas Law	$P - k_b T_g \sum_{i=1}^{11} n_i = 0$	Thermodynamic equilibrium between gas pressure P, temperature T_g, and heavy species n_i.
Quasi-Neutrality	$n_e + n_{\text{neg\_ions}} - n_{\text{pos\_ions}} = 0$	Local charge neutrality balancing electrons and ions.
Current	$I - e\, n_e\, v_d\, \pi R^2 = 0$	Couples discharge current I, electron density n_e, drift velocity v_d, and tube radius R.

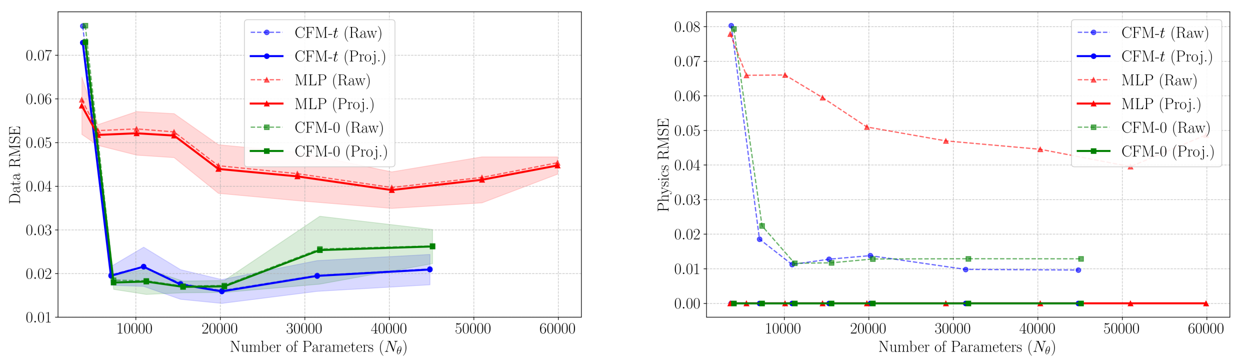
4.2 Results and Discussion

To comprehensively evaluate the proposed continuous-time generative frameworks, we assess model performance across three dimensions: model capacity (neural scaling), data efficiency, and missing value imputation. We compare the time-dependent (CFM-t), autonomous (CFM-0) formulations and, a standard multi-layer perceptron (MLP) regressors.

Performance is quantified on a held-out test set using two primary metrics: data RMSE for state-variable accuracy and physics RMSE for governing-equation residual violations. To isolate the network's intrinsic geometric representation from explicit constraint enforcement, we evaluate both raw predictions (dashed lines) and projected states (solid lines). All reported metrics reflect the mean and standard deviation computed across 5 independent random seeds for networks' initialization and training.

4.2.1 Model Capacity Scaling

In the zero-shot surrogate modeling task, no variables are observed at inference ($k = 0$), requiring the models to map the state y from the prior distribution. Figures 1a and 1b display the data and physics RMSE as a function of the number of trainable parameters (N_θ).



(a) Data RMSE

(b) Physics RMSE

Figure 1: Model capacity scaling on the default LTP dataset for the pure surrogate modeling task ($k = 0$). Both generative frameworks (CFM-t and CFM-0) outperform the baseline MLP across parameter counts, achieving lower data RMSE while inherently preserving lower physical constraint violations prior to post-hoc projection.

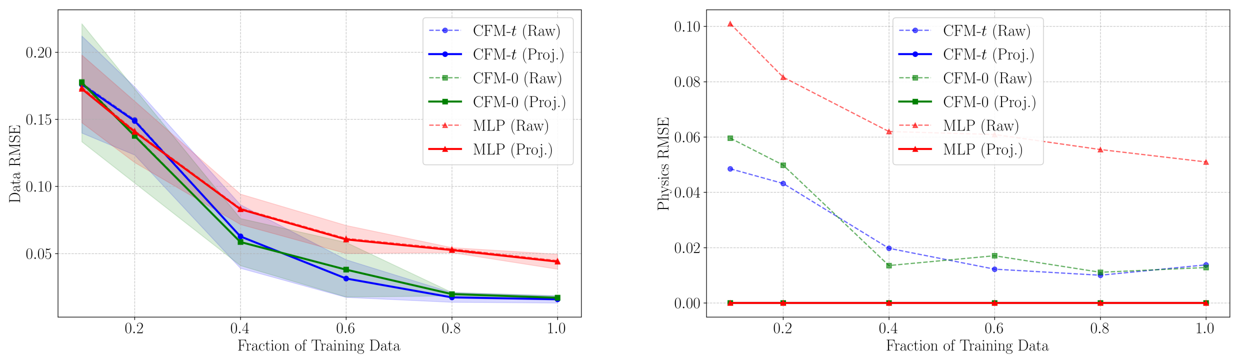
Both generative Flow Matching architectures (CFM-t and CFM-0) consistently outperform the baseline MLP. While the CFMs achieve a projected data RMSE of approximately 0.020 using 19.5k parameters, the MLP saturates at a significantly higher error (≈ 0.040) even when its capacity is inflated to over 50k parameters. This performance gap validates that the continuous spatial trajectory resolved by the Flow Matching objective constitutes a superior structural prior for surrogate modeling compared to direct one-step regression. Furthermore, to confirm that this advantage stems specifically from targeting the clean physical state, a comprehensive ablation study comparing y-prediction against standard velocity v-prediction is provided in Appendix B, demonstrating that y-prediction is required for parameter-efficient learning.

The physics RMSE results highlight another critical advantage. While the projection step enforces global constraints by definition, driving the Physics RMSE to machine zero for all projected states, the raw predictions reveal differences in learned geometry. Raw MLP outputs exhibit substantial constraint residuals, whereas both raw CFM models achieve lower physical errors prior to projection. This confirms that the y-prediction flow inherently captures the deterministic geometry of the physical manifold $\mathcal{M}$, reducing reliance on post-hoc corrections.

Moreover, the time-independent CFM-0 achieves accuracy on par with the time-dependent CFM-t across all parameter scales, validating that the model learns a generative transport without requiring a temporal clock.

4.3 Robustness to Data Scarcity

High-fidelity physical data is often limited by sensor costs or simulation constraints. To evaluate sample efficiency, we trained the default models ($N_\theta \approx 20$k) on varying fractions of the training set while evaluating performance on the fixed, full-size test set.



(a) Data RMSE (b) Physics RMSE

Figure 2: Sample efficiency evaluated on the fixed test set across varying training data fractions ($N_\theta \approx 20$k). The generative models (CFM-t and CFM-0) maintain lower data RMSE (a) and stable pre-projection physics RMSE (b) under data scarcity, whereas the baseline MLP degrades faster.

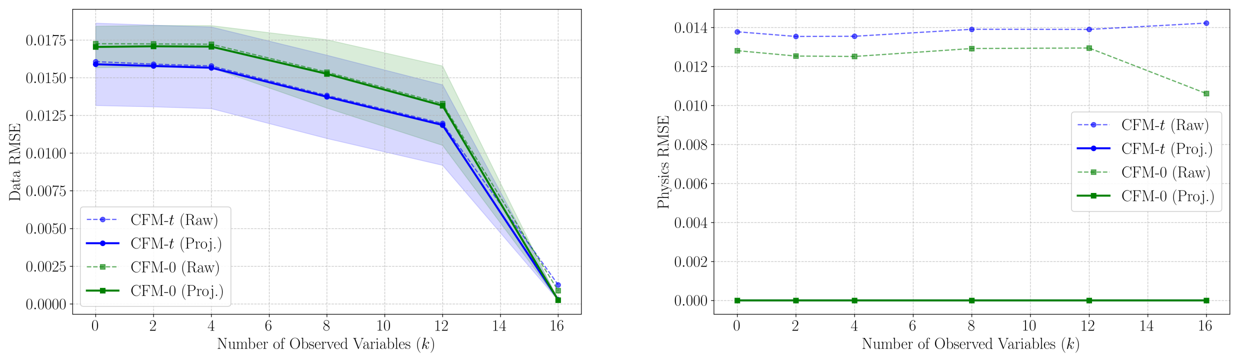
As shown in Figure 2, the performance gap between the generative models and the baseline MLP widens under data scarcity. With 50% of the training data, the CFM models maintain a Data RMSE of ≈ 0.05, whereas the MLP degrades substantially (Figure 2a). Furthermore, the CFMs' raw physical consistency remains low and stable across the majority of the sample sizes, while the MLP's Physics RMSE deteriorates faster as data decreases (Figure 2b). This sample efficiency stems directly from the data-dependent interpolant, which inherently exploits the low-dimensional geometry of the physical manifold.

4.4 Progressive Missing Data Imputation

Finally, we transition from the pure surrogate task ($k = 0$) to missing data imputation ($k > 0$). Using the exact same trained weights, we supply the network with an increasing number of known physical measurements k to act as structural anchors.

As shown in Figure 3, the data RMSE monotonically decays for both models as the number of observations increases, approaching zero when the state is nearly fully observed ($k = 16$). While both models excel at data assimilation, their measurement-locking mechanisms differ. CFM-t requires continuous boundary corrections, which rely on evaluating the exact temporal noise schedule σ_t at every ODE solver increment. Conversely, CFM-0 achieves identical accuracy through a simpler update: replacing the observed dimensions with the clean anchors at each fixed-point iteration.

Notably, across all experimental configurations, the difference between the raw and projected data RMSE remains negligibly small. While the hard projection strictly enforces global conservation laws without degrading predictive accuracy, it provides no substantial reduction in Euclidean distance to the ground truth. This empirical behavior directly validates our earlier geometric analysis (Section 3.4): the physical constraint space $\mathcal{C}_{\text{obs}}$ is vastly higher-dimensional than the intrinsic data manifold $\mathcal{M}_{\text{obs}}$. Consequently, merely satisfying macroscopic equations does not guide predictions closer to the true physical state, underscoring the necessity of a strong, data-driven generative prior.



(a) Data RMSE

(b) Physics RMSE

Figure 3: Progressive missing data imputation (data assimilation). As the number of observed physical anchors (k) increases, the state reconstruction error decays. The autonomous formulation (CFM-0) achieves parity with the time-dependent formulation (CFM-t) while utilizing a simpler boundary enforcement process.

5 Conclusion

In this work, we introduced a unified continuous-time conditional Flow Matching formulation for steady-state physical surrogate modeling and sparse data assimilation, deployed on a low-temperature plasma system. By integrating data-dependent stochastic interpolants with a dynamic masking operator, our formulation unifies surrogate prediction and progressive missing data imputation into a single generative formulation. Crucially, transitioning from standard velocity matching to a direct y-prediction objective resolves the zero conditional variance of deterministic physics, allowing the network to learn low-dimensional manifold representations with superior parameter efficiency and sample robustness compared to classical MLP baselines and stochastic transport models.

In addition, the formulation provides flexible inference pathways, accommodating both standard time-dependent integration (CFM-t) and an autonomous, clock-free transport regime (CFM-0) that enables exact measurement conditioning through simple anchor substitution without boundary disruption. By decoupling manifold representation learning from exact post-hoc geometric projections, we ensure global conservation satisfaction without introducing optimization instabilities from soft penalty losses. Our results also reveal that satisfying macroscopic conservation laws alone provides negligible corrections toward the ground truth, underscoring that global constraints are high-dimensional relative to the true data manifold and establishing the necessity of a strong generative prior.

Future work will extend this formulation by scaling the architecture to higher-dimensional spatial domains and hybrid datasets that pair sparse experimental measurements with numerical simulations, leveraging dynamic masking and denoising as mechanisms for data augmentation and improved sample efficiency. Ultimately, these findings indicate that implicitly learning the geometric manifold of steady-state physical systems provides a scalable, robust, and data-efficient alternative when explicit equation-based formulations are unavailable or computationally intractable.

Code Availability

The source code and scripts used to produce the results in this paper are publicly available and can be accessed at: [CHANGE THE LINK] https://github.com/joseAf28/CDM-PhysicsSurrogate.

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References

[1] M. Raissi, P. Perdikaris, and G.E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[2] George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3(6):422–440, 2021.

[3] Matilde Valente, Tiago C. Dias, Vasco Guerra, and Rodrigo Ventura. Physics-consistent machine learning with output projection onto physical manifolds. *Communications Physics*, 8(1):433, 2025.

[4] Tiago C Dias, Chloé Fromentin, Luís L Alves, Antonio Tejero-del Caz, Tiago Silva, and Vasco Guerra. A reaction mechanism for oxygen plasmas. *Plasma Sources Science and Technology*, 32(8):084003, aug 2023.

[5] Pedro Viegas, Jorge Silveira, Tiago Cunha Dias, Olivier Guaitella, Ana Sofía Morillo Candás, and Vasco Guerra. Surface recombination in pyrex in oxygen dc glow discharges: mesoscopic modelling and comparison with experiments. *Plasma Sources Science and Technology*, 33(5):055003, 2024.

[6] Mark K. Transtrum, Benjamin B. Machta, and James P. Sethna. Geometry of nonlinear least squares with applications to sloppy models and optimization. *Phys. Rev. E*, 83:036701, Mar 2011.

[7] Anuj Karpatne, Gowtham Atluri, James H. Faghmous, Michael Steinbach, Arindam Banerjee, Auroop Ganguly, Shashi Shekhar, Nagiza Samatova, and Vipin Kumar. Theory-guided data science: A new paradigm for scientific discovery from data. *IEEE Transactions on Knowledge and Data Engineering*, 29(10):2318–2331, October 2017.

[8] Tom Beucler, Michael Pritchard, Stephan Rasp, Jordan Ott, Pierre Baldi, and Pierre Gentine. Enforcing analytic constraints in neural networks emulating physical systems. *Physical Review Letters*, 126(9), March 2021.

[9] Yuntian Chen, Dou Huang, Dongxiao Zhang, Junsheng Zeng, Nanzhe Wang, Haoran Zhang, and Jinyue Yan. Theory-guided hard constraint projection (hcp): A knowledge-based data-driven scientific machine learning method. *Journal of Computational Physics*, 445:110624, November 2021.

[10] Antonio Tejero-del Caz, Vasco Guerra, Duarte Gonçalves, Mário Lino da Silva, Luís Marques, Nuno Pinhão, Carlos D Pintassilgo, and Luís L Alves. The Lisbon KInetics Boltzmann solver. *Plasma Sources Science and Technology*, 28(4):043001, 2019.

[11] L. L. Alves, A. Tejero-Del-Caz, L. Marques, P. Pereira, N. Pinhão, C. D. Pintassilgo, T. Silva, P. Viegas, and V. Guerra. Loki-gm: a global model framework for plasma chemistry studies, 2026.

[12] Charles K. Birdsall and A. Bruce Langdon. *Plasma Physics via Computer Simulation*. McGraw-Hill, New York, 1985.

[13] Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, May 2020.

[14] Alexander Lavin, David Krakauer, Hector Zenil, Justin Gottschlich, Tim Mattson, Johann Brehmer, Anima Anandkumar, Sanjay Choudry, Kamil Rocki, Atılım Güneş Baydin, Carina Prunkl, Brooks Paige, Olexandr Isayev, Erik Peterson, Peter L. McMahon, Jakob Macke, Kyle Cranmer, Jiaxin Zhang, Haruko Wainwright, Adi Hanuka, Manuela Veloso, Samuel Assefa, Stephan Zheng, and Avi Pfeffer. Simulation intelligence: Towards a new generation of scientific methods, 2022.

[15] Pratik Ghosh, Bhaskar Chaudhury, Shishir Purohit, Vishv Joshi, Ashray Kothari, and Devdeep Shetranjiwala. Deep learning assisted microwave-plasma interaction based technique for plasma density estimation. *Journal of Physics D: Applied Physics*, 57(1):014001, October 2023.

[16] Jiequn Han, Chao Ma, Zheng Ma, and Weinan E. Uniformly accurate machine learning-based hydrodynamic models for kinetic equations. *Proceedings of the National Academy of Sciences*, 116(44):21983–21991, October 2019.

[17] Benjamin Peherstorfer, Karen Willcox, and Max Gunzburger. Survey of multifidelity methods in uncertainty propagation, inference, and optimization, 2018.

[18] Alexander I.J. Forrester and Andy J. Keane. Recent advances in surrogate-based optimization. *Progress in Aerospace Sciences*, 45(1):50–79, 2009.

[19] Philipp Holl, Vladlen Koltun, and Nils Thuerey. Learning to control pdes with differentiable physics, 2020.

[20] Salvatore Cuomo, Vincenzo Schiano di Cola, Fabio Giampaolo, Gianluigi Rozza, Maziar Raissi, and Francesco Piccialli. Scientific machine learning through physics-informed neural networks: Where we are and what's next, 2022.

[21] Sifan Wang, Yujun Teng, and Paris Perdikaris. Understanding and mitigating gradient pathologies in physics-informed neural networks, 2020.

[22] Aditi S. Krishnapriyan, Amir Gholami, Shandian Zhe, Robert M. Kirby, and Michael W. Mahoney. Characterizing possible failure modes in physics-informed neural networks, 2021.

[23] Sifan Wang, Shyam Sankaran, and Paris Perdikaris. Respecting causality is all you need for training physics-informed neural networks, 2022.

[24] Arka Daw, Jialiang Bu, Sifan Wang, Paris Perdikaris, and Anuj Karpatne. Rethinking the importance of sampling in physics-informed neural networks. *arXiv preprint arXiv:2207.02338*, 2022.

[25] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *arXiv preprint arXiv:2006.11239*, 2020.

[26] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *arXiv preprint arXiv:2011.13456*, 2021.

[27] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling, 2023.

[28] Michael S. Albergo and Eric Vanden-Eijnden. Building normalizing flows with stochastic interpolants, 2023.

[29] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022.

[30] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *arXiv preprint arXiv:2303.04137*, 2023.

[31] Zhengyang Geng, Mingyang Deng, Xingjian Bai, J. Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling, 2025.

[32] Jan-Hendrik Bastek, WaiChing Sun, and Dennis M. Kochmann. Physics-informed diffusion models, 2025.

[33] Zhao Wei, Chin Chun Ooi, Abhishek Gupta, Jian Cheng Wong, Pao-Hsiung Chiu, Sheares Xue Wen Toh, and Yew-Soon Ong. Evolvable conditional diffusion, 2025.

[34] Hyungjin Chung, Jeongsol Kim, Michael T. Mccann, Marc L. Klasky, and Jong Chul Ye. Diffusion posterior sampling for general noisy inverse problems, 2024.

[35] Yankun Hong, Harshit Bansal, and Karen Veroy. A score-based generative solver for pde-constrained inverse problems with complex priors. *arXiv preprint arXiv:2408.16626*, 2024.

[36] Christian Jacobsen, Yilin Zhuang, and Karthik Duraisamy. Cocogen: Physically consistent and conditioned score-based generative models for forward and inverse problems. *SIAM Journal on Scientific Computing*, 47(2):C399–C425, 2025.

[37] Alexander Tong, Kilian Fatras, Nikolay Malkin, Guillaume Huguet, Yanlei Zhang, Jarrid Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport, 2024.

[38] Giacomo Baldan, Qiang Liu, Alberto Guardone, and Nils Thuerey. Physics vs distributions: Pareto optimal flow matching with physics constraints, 2026.

[39] Hanru Bai, Yuncheng Zhou, and Difan Zou. Physics-informed neural pde solvers via spatio-temporal mean-flow. In *Proceedings of the AI4Physics Workshop at the 43rd International Conference on Machine Learning (AI4Physics@ICML 2026)*, Seoul, South Korea, 2026.

[40] Michael S. Albergo, Mark Goldstein, Nicholas M. Boffi, Rajesh Ranganath, and Eric Vanden-Eijnden. Stochastic interpolants with data-dependent couplings, 2024.

[41] Ségolène Martin, Anne Gagneux, Paul Hagemann, and Gabriele Steidl. Pnp-flow: Plug-and-play image restoration with flow matching, 2025.

[42] Chenrui Ma, Xi Xiao, Tianyang Wang, Xiao Wang, and Yanning Shen. Learning straight flows: Variational flow matching for efficient generation, 2025.

[43] Pascal Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7):1661–1674, 2011.

[44] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners, 2021.

[45] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems*, volume 35, pages 26565–26577, 2022.

[46] Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models, 2022.

[47] Tianhong Li and Kaiming He. Back to basics: Let denoising generative models denoise. *arXiv preprint arXiv:2511.13720*, 2025.

[48] Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. Extracting and composing robust features with denoising autoencoders. *Proceedings of the 25th International Conference on Machine Learning*, pages 1096–1103, 2008.

[49] Lee Cheuk-Kit, Florentin Coeurdoux, Peter Potaptchik, Yilun Du, Michael Samuel Albergo, and Eric Vanden-Eijnden. Beckmann transport models: From autonomous flows to one-step maps, 2026.

[50] José Afonso, Vasco Guerra, and Pedro Viegas. Stochastic hierarchical data-driven optimization: application to plasma-surface kinetics. *Machine Learning: Science and Technology*, 7(2):025024, mar 2026.

[51] Mark K. Transtrum, Benjamin B. Machta, and James P. Sethna. Why are nonlinear fits to data so challenging? *Phys. Rev. Lett.*, 104:060201, Feb 2010.

[52] Hugues Van Assel, Mark Ibrahim, Tommaso Biancalani, Aviv Regev, and Randall Balestriero. Joint-embedding vs reconstruction: Provable benefits of latent space prediction for self-supervised learning. In D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen, editors, *Advances in Neural Information Processing Systems*, volume 38, pages 21897–21937. Curran Associates, Inc., 2025.

[53] Youngkyu Kim, Youngsoo Choi, David Widemann, and Tarek Zohdi. A fast and accurate physics-informed neural network reduced order model with shallow masked autoencoder. *Journal of Computational Physics*, 451:110841, February 2022.

[54] Zuzanna Gawrysiak and Krzysztof Krawiec. Physics-informed spectral modeling for hyperspectral imaging. *IEEE Geoscience and Remote Sensing Letters*, 23:1–5, 2026.

[55] Guillaume Alain andYoshua Bengio. What regularized auto-encoders learn from the data generating distribution. *arXiv preprint arXiv:1211.4246*, 2014.

[56] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations, 2019.

[57] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow, 2022.

[58] Aram-Alexandre Pooladian, Heli Ben-Hamu, Carles Domingo-Enrich, Brandon Amos, Yaron Lipman, and Ricky T. Q. Chen. Multisample flow matching: Straightening flows with minibatch couplings, 2023.

[59] Andreas Lugmayr, Martin Danelljan, Andres Romero, Fisher Yu, Radu Timofte, and Luc Van Gool. Repaint: Inpainting using denoising diffusion probabilistic models, 2022.

[60] Runqian Wang and Yilun Du. Equilibrium matching: Generative modeling with implicit energy-based models, 2025.

[61] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 5998–6008, 2017.

[62] Brian L Trippe, Jason Yim, Doug Tischer, David Baker, Tamara Broderick, Regina Barzilay, and Tommi Jaakkola. Diffusion probabilistic modeling of protein backbones in 3d for the motif-scaffolding problem. *arXiv preprint arXiv:2206.04119*, 2022.

[63] Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J. Fleet. Video diffusion models. *arXiv preprint arXiv:2204.03458*, 2022.

[64] Arne Schneuing, Yuanqi Du, Charles Harris, Arian Jamasb, Ignavier Ng, Weitao Du, Carla P. Gomes, Christian Blundell, Pietro Lio, Tommi Jaakkola, Nathan Cho, and Tom Blundell. Structure-based drug design with equivariant diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 37535–37549, 2022.

[65] Liang Wang, Chao Song, Zhiyuan Liu, Yu Rong, Qiang Liu, Shu Wu, and Liang Wang. Diffusion models for molecules: A survey of methods and tasks. *arXiv preprint arXiv:2502.09511*, 2025.

[66] Sifan Wang, Yujan Teng, and Paris Perdikaris. Understanding and mitigating gradient pathologies in physics-informed neural networks. *arXiv preprint arXiv:2001.04536*, 2020.

[67] Steven L. Brunton and J. Nathan Kutz. *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control*. Cambridge University Press, 2019.

[68] Tianyi Zeng, Tianyi Wang, Jiaru Zhang, Zimo Zeng, Feiyang Zhang, Yiming Xu, Sikai Chen, Junfeng Jiao, Christian Claudel, and Xinbo Chen. Pild: Physics-informed learning via diffusion, 2026.

[69] Miranda J. S. Horne, Peter K. Jimack, Amirul Khan, and He Wang. Hard constraint projection in a physics informed neural network. *Proceedings of the 35th Parallel CFD International Conference 2024 35th Parallel CFD International Conference 2024*, ParCFD 2024:50–53 (2025). doi:10.34734/FZJ, 2025.

[70] Antonio Tejero-del Caz, Vasco Guerra, Nuno Pinhão, Carlos D Pintassilgo, and Luís L Alves. On the quasi-stationary approach to solve the electron boltzmann equation in pulsed plasmas. *Plasma Sources Science and Technology*, 30(6):065008, 2021.

[71] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.

[72] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4195–4205, 2023.

[73] Joel A. E. Andersson, Joris Gillis, Greg Horn, James B. Rawlings, and Moritz Diehl. CasADi: a software framework for nonlinear optimization and optimal control. *Mathematical Programming Computation*, 11(1):1–36, 2019.

[74] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization, 2019.

[75] Ilya Loshchilov and Frank Hutter. Sgdr: Stochastic gradient descent with warm restarts, 2017.

Appendix

A Architectures and Experimental Setup

To ensure full reproducibility, this section details the specific neural network architectures, training procedures, and default hyperparameters utilized across all experiments.

A.1 Neural Network Architectures

The continuous-time generative models (CFM-t and CFM-0) are parameterized using an Adaptive Layer Normalization (AdaLN) residual architecture. The input state I_t and the binary observation mask S are concatenated in the network and projected into the hidden space. The exact structural configurations, tailored to match a constrained budget of ≈ 19.5k parameters, are summarized in Table 2.

Table 2: Default Neural Network Architectures ($N_\theta \approx 19500$)

Component	**CFM-t**	**CFM-0**
Input Dimension (x)	3	3
Output Dimension (y)	17	17
Hidden Dimension	40	40
Conditioning Dimension	32	32
Residual Blocks	2	2
Temporal Embedding	Sinusoidal	None
Activation Function	SiLU	SiLU

A.2 Training Procedure and Hyperparameters

Both models are trained utilizing a batch size of 64 and an initial learning rate of 0.001. Weights are optimized using the AdamW optimizer with a weight decay of 10^{-4} [74]. To stabilize the learned continuous vector fields, we apply an Exponential Moving Average (EMA) to the network weights with a decay factor of 0.999 [25, 26]. The learning rate is decayed following a Cosine Annealing schedule [75].

To ensure robust convergence, the models are trained for a minimum of 5000 epochs and a maximum of 6000 epochs, monitored via an early stopping criterion with a patience of 4 validation intervals. The complete set of default hyperparameters is presented in Table 3.

Table 3: Default Training and Inference Hyperparameters

Hyperparameter	**Value**
Batch Size	64
Initial Learning Rate	0.001
Optimizer	AdamW
Weight Decay	10^{-4}
Learning Rate Scheduler	Cosine Annealing
EMA Decay	0.999
Minimum Epochs	5000
Maximum Epochs	6000
Prediction Target	y-prediction
Maximum Noise (σ)	1.0
ODE Solver (CFM-t)	Heun
Integration Steps (CFM-t)	100
Max Integration Steps (CFM-0)	200
Virtual step size η (CFM-0)	0.1
Convergence threshold ϵ_{conv} (CFM-0)	10^{-7}

B Ablation Studies

B.1 Target Objective: Parameter Efficiency of y-Prediction vs. v-Prediction

As established in our theoretical formulation, the standard velocity-prediction (v-prediction) paradigm targets an instantaneous vector field that retains explicit dependence on the ambient noise variable ϵ. For deterministic scientific systems characterized by zero conditional variance ($\text{Var}[y|x] = 0$), forcing the network to regress this stochastic target is statistically less efficient. Conversely, predicting the clean physical state (y-prediction) targets the deterministic invariant manifold directly, theoretically requiring significantly less network capacity to resolve.

To empirically validate this structural advantage, we evaluated both paradigms across a sweeping range of model capacities, scaling from $N_\theta \approx 5$k to 45k parameters.

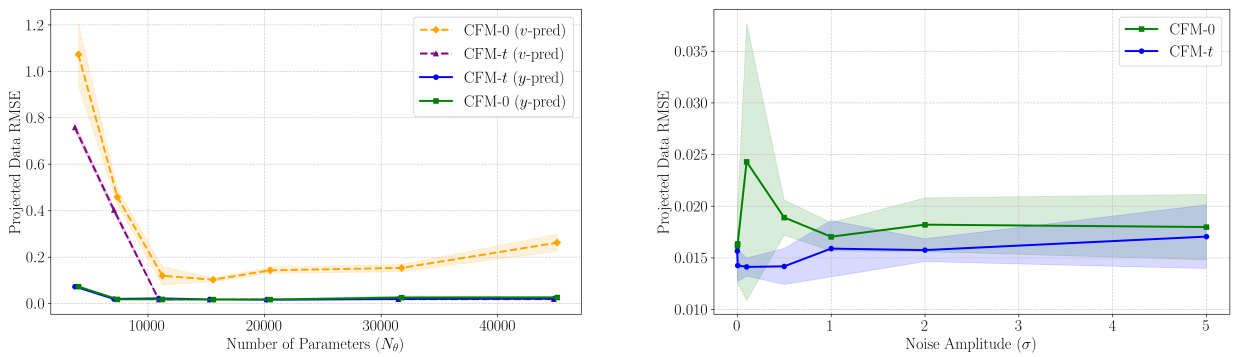
As presented in Figure 4a, at higher capacities, both objectives converge to comparable baseline errors. However, as the network capacity is constrained, the v-prediction models (dashed lines) suffer performance degradation. Specifically, the error for the v-prediction CFM model spikes at lower parameter counts. This indicates that limited-capacity networks lack the expressive power to effectively average over the high-variance noise distribution and recover the underlying geometric flow.

In contrast, the y-prediction models (solid lines) remain stable and competitive even at the lowest parameter extremes. By forcing the ambient noise to act merely as a spatial probe to construct the manifold geometry, rather than serving as a direct regression target, y-prediction proves to be a vastly more scalable, robust, and parameter-efficient paradigm for continuous-time physical surrogate modeling.

B.2 Influence of Noise Amplitude (σ)

A critical component of our data-dependent interpolant is the injection of isotropic Gaussian noise. As discussed in the methodology, scaling this noise via σ provides essential manifold smoothing, ensuring the base density is well-defined over the entire higher-dimensional ambient space and preventing out-of-distribution changes during training.

To understand the sensitivity of the models to the boundary of this ambient space, we varied σ from near-zero (10^{-7}) up to 5.0. The resulting projected Data RMSE is shown in Figure 4b. It reveals distinct stability profiles between the two formulations governed by their temporal dynamics. For CFM-t, small noise amplitudes ($\sigma \in [0.1, 0.5]$) yield the lowest error and minimal variance by shortening the transport path between prior and target distributions, while explicit time indexing maintains stable integration [40]. In contrast, CFM-0 exhibits acute sensitivity and high variance in the small-noise regime ($\sigma \approx 0.1$) due to its reliance on a time-independent vector field, where small perturbations disrupt autonomous fixed-point contraction. As noise increases ($\sigma \geq 1.0$), CFM-0 stabilizes near RMSE ≈ 0.018 as the asymmetric gating operator provides sufficient smoothing over unobserved dimensions. Setting $\sigma \in [0.5, 1.0]$ provides a good balance, ensuring stable convergence for CFM-0 while preserving near-peak accuracy for CFM-t.



(a) Target Objective (y-pred vs. v-pred)

(b) Noise Amplitude (σ)

Figure 4: Ablation analysis on target objectives and noise hyperparameters. (a) Comparison of target parameterizations across model capacity: direct y-prediction consistently outperforms velocity matching (v-prediction), particularly in low-capacity regimes. (b) Sensitivity to noise amplitude σ, demonstrating that both CFM formulations maintain stable predictive accuracy across orders of magnitude of ambient noise.

C Algorithmic Implementations and Pseudo-Code

This section provides the implementation details and pseudo-code for training and inference across the proposed Conditional Flow Matching formulations. Algorithm 1 outlines the unified training objective utilizing the data-dependent interpolant with dynamic masking and y-prediction. Algorithms 2 and 3 detail the respective inference routines: time-dependent numerical integration for CFM-t and the clock-free, fixed-point transport regime for CFM-0.

Algorithm 1 CFM Surrogate Training

Input : Dataset $p_d(x, y)$, Schedule $(\alpha_t, \beta_t, \sigma_t)$, Mask distribution $p(S)$, Network ϕ
Output : Trained Network ϕ
while *not converged* **do**
  ▷ *Sample Data, Time, & Mask*:
  $x, y \sim p_d(x, y), \quad t \sim \mathcal{U}(0, 1), \quad S \sim p(S)$
  ▷ *Sample Ambient Noise*:
  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$
  ▷ *Construct Generalized Interpolant*:
  if *CFM-t (Time-Dependent)* **then**
    $I_t \leftarrow S \odot (y + \sigma_t \epsilon) + (\mathbf{1} - S) \odot (\beta_t y + \sigma_t \epsilon)$
    $\hat{y} \leftarrow g_\phi(I_t, x, S, t)$
  else
    $I_t \leftarrow S \odot y + (\mathbf{1} - S) \odot (\beta_t y + \sigma_t \epsilon)$ ▷ *CFM-0*
    $\hat{y} \leftarrow g_\phi(I_t, x, S)$
  end
  ▷ *Compute y-prediction Loss*:
  $\mathcal{L}_y(\phi) \leftarrow \frac{1}{B} \sum_{i=1}^{B} \|\hat{y}_i - y_i\|_2^2$
  ▷ *Update Parameter Space*:
  $\phi \leftarrow \phi - \alpha \nabla_\phi \mathcal{L}_y(\phi)$ ▷ *AdamW Update*
end

Algorithm 2 CFM-t Continuous Inference

Input : Condition x, Sensor Obs y_{obs}, Mask S, Model g_ϕ, Steps N, Euler Solver
▷ *Sample and fix global noise realization*:
$\epsilon \sim \mathcal{N}(0, \mathbf{I}), \quad dt \leftarrow 1/N$
$I_0 \leftarrow S \odot (y_{\text{obs}} + \sigma_0 \epsilon) + (\mathbf{1} - S) \odot (\sigma_0 \epsilon)$
for $i = 0$ ***to*** $N - 1$ **do**
  $t \leftarrow i \cdot dt$
  ▷ *Predict clean physical state*:
  $\hat{y} \leftarrow g_\phi(I_t, x, S, t)$
  ▷ *Convert to instantaneous ODE velocity*:
  $\hat{v}_t \leftarrow \frac{\dot{\alpha}_t}{\alpha_t} I_t + \left(\dot{\beta}_t - \frac{\dot{\alpha}_t \beta_t}{\alpha_t}\right) \hat{y}$
  ▷ *Solver Step & Strict Harmonization*:
  $I_{t+dt}^{\text{raw}} \leftarrow I_t + \hat{v}_t, dt$ ▷ *Euler Update*
  $I_{t+dt} \leftarrow S \odot (y_{\text{obs}} + \sigma_{t+dt} \epsilon) + (\mathbf{1} - S) \odot I_{t+dt}^{\text{raw}}$
end
▷ *Hard Projection onto Physical Invariants*:
$y^* \leftarrow \Pi_{\mathcal{C}}(I_1) := \arg\min_{y \in \mathcal{C}} \|I_1 - y\|_2^2$
return y^*

Algorithm 3 CFM-0 Autonomous Inference

Input : Condition x, Sensor Obs y_{obs}, Mask S, Model g_ϕ, Step size η
▷ *Asymmetric Initialization*:
$\epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad I^{(0)} \leftarrow S \odot y_{\text{obs}} + (\mathbf{1} - S) \odot (\sigma \epsilon)$
for $k = 0$ ***to*** $N_{\max}$ **do**
  ▷ *Predict clean physical manifold projection*:
  $\hat{y} \leftarrow g_\phi(I^{(k)}, x, S)$
  $v \leftarrow \hat{y} - I^{(k)}$
  if $\|(\mathbf{1} - S) \odot v\|_2 < \epsilon_{conv}$ **then**
    break
  end
  ▷ *Autonomous Fixed-Point Update & Harmonization*:
  $I_{\text{raw}}^{(k+1)} \leftarrow I^{(k)} + \eta\, v$
  $I^{(k+1)} \leftarrow S \odot y_{\text{obs}} + (\mathbf{1} - S) \odot I_{\text{raw}}^{(k+1)}$
end
▷ *Hard Projection onto Physical Invariants*:
$y^* \leftarrow \Pi_{\mathcal{C}}(I^{(k+1)}) := \arg\min_{y \in \mathcal{C}} \|I^{(k+1)} - y\|_2^2$
return y^*